
OpenSTREAM Manual

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WELCOME

OpenSTREAM is a MATLAB program for simulating two-phase flow behaviors, inspired by Le Corre [2022].

Add more descriptive text.

The purpose of this documentation is to fully describe the usage and thought-process behind the design of the program.

The PDF version of this manual can be found here: [OpenSTREAM.pdf](#)

2.1 Getting started

2.1.1 Prerequisites and installation

OpenSTREAM is a MATLAB program for simulating two-phase flow behaviors. The Python version of [CoolProp](#) is used for calculating the fluid properties. [CoolPropWrapper](#) is a MATLAB interface that interacts with CoolProp through Python. Both OpenSTREAM and CoolProp are hosted here on GitHub. Making a copy (cloning) of these programs will be our first step.

Install using Git and Github

- There are two main ways of interacting with GitHub to use and contribute to the OpenSTREAM project: [Git Command Line Interface \(CLI\)](#), and the [GitHub Desktop](#) program. Use the links to set up one of the methods.
- After installing and setting up your Git mechanism of choice, if you chose to use the:
 - **Command line interface**
 - * Clone (make a copy of) the **repository** using: `git clone --recursive git@github.com:OpenSTREAM-solvers/openstream.git` ([SSH authentication](#))
 - or
 - `git clone --recursive https://github.com/OpenSTREAM-solvers/openstream.git` ([HTTPS authentication](#)).
 - (The `--recursive` flag is needed to clone [submodules](#), such as the [mfval/CoolPropWrapper](#), in one command.)
 - * Then, use `cd openstream` to enter the newly created folder. You should see the newly downloaded files for the project.
 - **GitHub Desktop program**
 - * Open the program, login, and **Clone repository**. Everything should be cloned automatically. This is part of the setup guide linked above.

MATLAB-Python compatibility

Before using the OpenSTREAM, we need to make sure Python with the appropriate version is installed, and MATLAB knows the location of the Python executable. The following instructions are from the [CoolPropWrapper](#) repo.

1. Determine which versions of Python are compatible with your version of MATLAB: [compatibility list](#). The second column, *MATLAB Interface MATLAB Engine*, shows the compatible Python versions.
2. Start MATLAB and verify a compatible python version is installed:
 - Run `pyenv()`
 - If the results are empty or a incompatible Python version is shown, go to step 3.
 - If the results are satisfactory, continue to step 5.
3. Install compatible Python3
 - **Windows:** Go to [python.org](#) to download the specific version of Python you need. Follow the setup procedure.
 - **Linux:** Chances are, you already have a version of Python installed. Verify the version using `python3 --version`. If Python is not installed, or a incompatible version is installed, install appropriate version using the OS package manager. For Debian-based Linux distros (such as Ubuntu), use `sudo apt-get install python3.x` to install the *x* version of Python3.
 - **MacOS:** Someone with a Mac, please write how you do this...
4. Specify Python installation location in MATLAB:
 - Run `pyenv('Version', pathtopython')`, where `pathtopython` is the path to where Python is installed. Here are some typical locations depending on your OS:
 - **Windows:** `C:\Users\Username\AppData\Local\Programs\Python\Python310\python.exe`, for version 3.10.
 - **Linux:** `/usr/bin/python3.10`, for version 3.10. Use `whereis python3` to find possible locations.
 - **MacOS:** Someone with a Mac, please write how you do this...
5. Try creating an instance of `CoolPropWrapper()` in MATLAB:
 - Run `cp=CoolPropWrapper()`.
 - If you are using the `CoolPropWrapper` as MATLAB Package in OpenSTREAM, use `cp=CoolPropWrapper.CoolPropWrapper()`.
 - If the `CoolProp` module is not installed, you will be prompted to do so automatically. If you prefer to do this manually, in your terminal, run `python3 -m pip install --user -U CoolProp`, replacing `python3` with the specific python path as necessary.

Next steps

Congrats! Hopefully, at this point, you have successfully made a copy of this repo and installed the required programs. Next, let's run the Tutorial script to make sure everything is working properly.

2.2 Run sample script

At this point, you are ready to run the sample script. You will see some of the basic usages of OpenSTREAM.

```
% Clear existing variables from workspace
clearvars

% Import some of the package classes
import Inputs.*
import Solvers.*
import Solvers.Mixture.*

% InputSet is used to create a collection of input settings
inputSet = InputSet( ...
    modelFilePath      = './inputs/models.inp', modelID      = 'BARCS', ...
    optionsFilePath    = './inputs/options.inp', optionsID   = 'STEADY', ...
    geometryFilePath   = './inputs/geom.inp', geometryID    = 'BARC', ...
    bcFilePath         = './inputs/bc_barb.inp', ...
    sessionParentDir   = fullfile(pwd,'outputs'), ...
    overwriteSessionFiles = true, ...
    LOGMODE            = 'BOTH');

% Create a mixture solver
mixSolver = MixtureSolver(inputSet);

% mixSolver is initialized at construction. Here, it is explicitly initialized for
↳ clarity.
mixSolver.initializeSolver();

% Solve (does not accept any argument)
mixSolver.solve();

% Axial and temporal plots
mixSolver.plotz(1); % Axial plot at 1st time step
mixSolver.plotz(mixSolver.NTIME); % Axial plot at final time step
mixSolver.plott(mixSolver.NZ) % Temporal plot at last spatial
↳ node
mixSolver.plott(mixSolver.NZ,'solveMode','STEADY') % Temporal plot at last spatial
↳ node of steady state solution
mixSolver.plotzt(mixSolver.NZ,"solveMode","TRANSIENT"); % Spatial-temporal map of
↳ transient solution at channel exit

% Save results
mixSolver.saveResults(saveFormat="MAT");
```

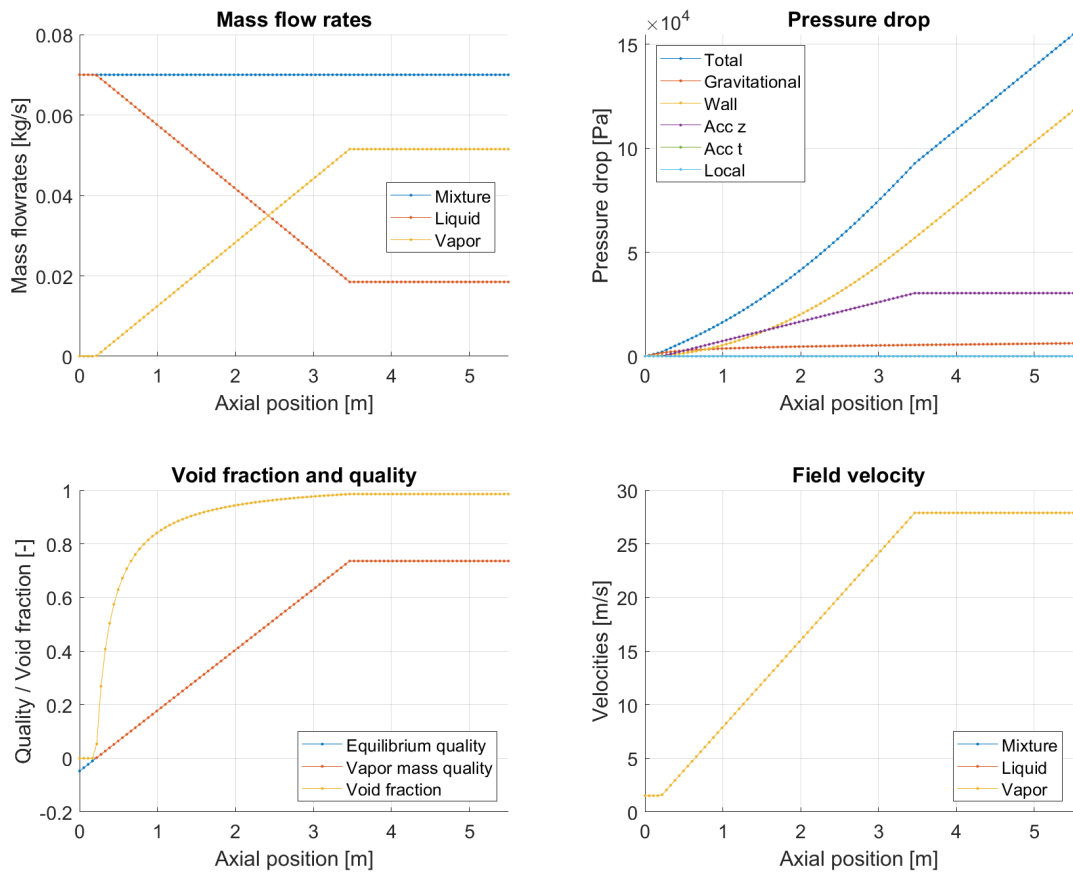


Fig. 1: Sample figure of axial distributions of mixture parameters at 0 [s].

PACKAGES

3.1 InputEnums

class InputEnums.**MOMENTBASE**

MOMENTBASE Base film momentum conservation model Defines the type of momentum conservation used on the liquid film. The conservation equations are used to solve for liquid-film velocity

class InputEnums.**FLUIDPROPERTIES**

FLUIDPROPERTIES Fluid property calculation assumptions Defines the pressure to be used for calculating fluid properties

class InputEnums.**LOCRELVEL**

LOCRELVEL Local relative velocity between liquid and vapor phase Defines the model for the local relative velocity

class InputEnums.**INTLENGTH**

INTLENGTH Interfacial length scale Defines the interfacial length scale model

class InputEnums.**MOMENTWAVE**

MOMENTWAVE Film momentum conservation model Defines the type of momentum conservation used on the liquid film. The conservation equations are used to solve for liquid-film velocity

class InputEnums.**SCBOIL**

SCBOIL Subcooled boiling model Detailed explanation goes here

class InputEnums.**WAVEFREQUENCY**

WAVEFREQUENCY Wave number conservation model This defines the type of conservation model used on the wave number in the Wave class.

class InputEnums.**THINFILMFRIC**

THINFILMFRIC Thin film friction model Detailed explanation goes here

class InputEnums.**TPKM**

TPKM Two-phase local loss multiplier Detailed explanation goes here

class InputEnums.**MOMENTDROP**

MOMENTDROP Drop momentum conservation model This defines the type of conservation model used on the droplets in the Drop class.

class InputEnums.**BOILCOEF**

BOILCOEF Wall boiling coefficient interpolation for subcooled boiling Detailed explanation goes here

class InputEnums.OAF

OAF Onset of annular flow model Defines the model to predict the position at which the onset of annular flow occurs based on local flow conditions. Wallis correlation is Equation 11.1019 in One-dimensional two-phase flow. Wallis (1969). The correlation was built using air-water data.

class InputEnums.INTAREA

INTAREA Interfacial area Defines the interfacial area model

class InputEnums.TPFM

TPFM Two-phase friction multiplier Detailed explanation goes here

class InputEnums.ENTRAINMENT

ENTRAINMENT Entrainment models Defines which droplet entrainment mass flux model is used. NOTE: the entrainment and deposition correlations are coupled, i.e. the same model should be used for both entrainment and deposition mass flux

class InputEnums.DROPDRAG

DROPDRAG drop drag model Detailed explanation goes here

class InputEnums.INTTRANSH

INTTRANSH Interfacial enthalpy transfer Defines the interfacial enthalpy exchange model from interfacial mass transfer The mass can be transferred either at phase bulk enthalpy, or at saturated enthalpy

class InputEnums.DEPOSITION

DEPOSITION Deposition models Defines which droplet deposition mass flux model is used. NOTE: the entrainment and deposition correlations are coupled, i.e. the same model should be used for both entrainment and deposition mass flux

class InputEnums.BASEEQTHICK

BASEEQTHICK Base film equilibrium thickness model Defines which base film equilibrium model to use

class InputEnums.OAFENTRAINED

OAFENTRAINED Entrained model at onset of annular flow Detailed explanation goes here

class InputEnums.MOMENTLIQUID

MOMENTLIQUID Liquid momentum conservation model This defines the type of conservation model used on the liquid phase in the liquid class.

class InputEnums.MOMENTGAS

MOMENTGAS Gas momentum conservation model This defines the type of conservation model used on the gas phase in the gas class.

class InputEnums.OAFFILMSPLIT

OAFFILMSPLIT Film mass flow ratio at onset of annular flow Detailed explanation goes here

class InputEnums.VOID

VOID Void fraction model This defines the type of void fraction model used. The homogenous model assumes the velocity of the fields is the same. The slip model assigns a slip ratio. The Bestion model is a correlation.

class InputEnums.MOMENTFILM

MOMENTFILM Film momentum conservation model Defines the type of momentum conservation used on the liquid film. The conservation equations are used to solve for liquid-film velocity

class InputEnums.VAPORFRIC

VAPORFRIC Vapor to film friction model Detailed explanation goes here

class InputEnums.INTNU

INTNU Interfacial Nusselt number Defines the interfacial Nusselt number model

class InputEnums.EQSTROUHAL

EQSTROUHAL Equilibrium Strouhal number Defines which base film equilibrium model to use

3.2 Inputs

Inputs is pretty cool.

class Inputs.BoundaryConditions(*filePath*, *geometryObjInput*)

Bases: *Inputs.Input*

BOUNDARYCONDITIONS Summary of this class goes here Detailed explanation goes here

Constructor Summary

BoundaryConditions(*filePath*, *geometryObjInput*)

BOUNDARYCONDITIONS Construct an instance of this class Detailed explanation goes here

Property Summary

TIME

Time [s]

PRESSURE = []

System pressure [Pa]

HIN = []

Inlet enthalpy [J/kg]

MFLOW = []

Mass flow rate [kg/s]

POWER = 0

Total power [W]

WMESH = 1

Relative power node size distribution [m]

WPOWER = 1

Relative power distribution(s) [-]

Method Summary

TSAT(*fluidObj*)

TSAT() [K] Saturation temperature given fluidObj

HF(*fluidObj*)

HF() [J/kg] Liquid saturation given fluidObj

HG(*fluidObj*)

HF [J/kg] Vapor saturation given fluidObj

TIN(*fluidObj*)

TIN() [K] Inlet temperature

DTIN(*fluidObj*)

DTIN() [K] Inlet subcooling temperature difference given fluidObj

DHIN(*fluidObj*)
 DHIN() [J/kg] Inlet subcooling enthalpy difference given fluidObj

XIN(*fluidObj*)
 XIN() [-] Inlet equilibrium quality

plot(*fluidObj*)
 PLOT Plot boundary conditions usage: bc.plot(fluidObj)

class Inputs.**InputSet**(*opts*)
 INPUTSET Creates set of input objects Detailed explanation goes here

Constructor Summary

InputSet(*opts*)
 INPUTSET Construct an instance of this class Detailed explanation goes here

class Inputs.**Options**(*filePath, optionsID*)
 Bases: *Inputs.Input*
 OPTIONS Summary of this class goes here Detailed explanation goes here

Constructor Summary

Options(*filePath, optionsID*)
 MODEL Construct an instance of this class Detailed explanation goes here

Property Summary

ID
 Option *ID*

TSTEP = 0.1
 Time step [s]

MAXITER = 100
 Max number of inner (point) iterations

ERRORW = 1E-3
 Mass flow rate error target in inner iterations [kg/s]

ERRORU = 1E-4
 Velocity error target in inner iterations [m/s]

ERRORP = 1E+0
 Pressure error target in inner iterations [Pa]

ERRORH = 1E+0
 Enthalpy error target in inner iterations [J/kg]

SSTSTEP = 1.0
 Time step for steady-state iterations [s]

SSMAXITER = 20
 Max number of steady-state iterations

SSCONVW = 1E-3
 Mass flow rate steady-state convergence criterion [kg/s]

SSCONVU = 1E-3

Velocity steady-state convergence criterion [m/s]

SSCONVP = 1E+0

Pressure steady-state convergence criterion [Pa]

SSCONVH = 1E+0

Enthalpy steady-state convergence criterion [J/kg]

AXIALINTERP = 'next'

Axial power interpolation method

TIMEINTERP = 'linear'

Time-dependant boundary conditions interpolation method

RELAXWM = 1

Relaxation factor for the mixture mass conservation equation

RELAXPM = 1

Relaxation factor for the mixture momentum conservation equation

RELAXHM = 1

Relaxation factor for the mixture energy conservation equation

RELAXWL = 1

Relaxation factor for the liquid mass conservation equation

RELAXWV = 1

Relaxation factor for the vapor mass conservation equation

RELAXUL = 1

Relaxation factor for the liquid momentum conservation equation

RELAXUV = 1

Relaxation factor for the vapor momentum conservation equation

RELAXHL = 1

Relaxation factor for the liquid mass conservation equation

RELAXHV = 1

Relaxation factor for the vapor energy conservation equation

ERRORWF = 1E-4

Film mass flow rate error target in inner iterations [kg/s/m]

ERRORUF = 1E-2

Film velocity error target in inner iterations [m/s]

ERRORUD = 1E-2

Drop velocity error target in inner iterations [m/s]

SSCONVWF = 1E-4

Film mass flow rate steady-state convergence criterion [kg/s/m]

SSCONVUF = 1E-2

Film velocity steady-state convergence criterion [m/s]

SSCONVUD = 1E-2

Drop velocity steady-state convergence criterion [m/s]

RELAXWF = 0.5

Relaxation factor for the film energy conservation equation

RELAXUF = 0.2

Relaxation factor for the film momentum conservation equation

RELAXUD = 0.2

Relaxation factor for the drop momentum conservation equation

RELAXWB = 0.5

Relaxation factor for the base mass conservation equation

RELAXUB = 0.2

Relaxation factor for the base momentum conservation equation

RELAXWW = 0.5

Relaxation factor for the wave mass conservation equation

RELAXUW = 0.2

Relaxation factor for the wave momentum conservation equation

RELAXFW = 0.5

Relaxation factor for the wave number conservation equation

class Inputs.**Model**(filePath, modelID)

Bases: [Inputs.Input](#)

MODEL Summary of this class goes here Detailed explanation goes here

Constructor Summary

Model(filePath, modelID)

MODEL Construct an instance of this class Detailed explanation goes here

Property Summary

ID

Model [ID](#)

NNODES = []

Number of axial nodes

FLUID = "WATER"

Fluid ID

PROPERTIES = 'SATURATED'

Fluid property assumptions

ANGLE = 0

Flow axis angle from vertical [deg]

FRICTION = [0.2-0.20]

Wall friction coefficients

TPFM = 'HOMOGENEOUS'
Two-phase friction multiplier [-]

KLOC = [00]
Elevation of local perturbations [m]

KLOSS = [00]
corresponding pressure loss coefficients [-]

TPKM = 'HOMOGENEOUS'
Two-phase local loss multiplier [-]

SCBOIL = 'NONE'
Subcooled boiling mode

VOID = 'HOMOGENEOUS'
Void fraction model

SLIP = 1
Phase velocity ratio [-]

MOMENTLIQUID = 'SLIP'
Liquid momentum conservation model [-]

MOMENTGAS = 'SLIP'
Gas momentum conservation model [-]

BOILCOEF = 'QUADRATIC'
Subcooled boiling interpolation [-]

INTLENGTH = 'CONSTANT'
Interfacial length scale model model [-]

INTAREA = 'DISPGAS2DISPLIQ'
Interfacial area model [-]

INTLENGTHCST = 2E-3
Constant interfacial length scale [m]

INTTRANSH = 'BULK'
Interfacial enthalpy transfer [-]

INTNU = 'CONSTANT'
Interfacial heat transfer model [-]

INTNUVCST = 2.0
Dispersed gas constant interfacial Nusselt number [-]

INTNULCST = 2.0
Dispersed liquid constant interfacial Nusselt number [-]

RANZMARSHALLVCST = [20.61/21/3]
Dispersed gas Ranz-Marshall coefficients [-]

RANZMARSHALLLCST = [20.61/21/3]
Dispersed liquid Ranz-Marshall coefficients [-]

LOCRELVEL = 'SCALED'

Local relative velocity model [-]

RELVELCST = 1E-1

Multiplication factor to determine the localrelative velocity [-]

RELAXTCOND = 0.2

Condensation relaxation time [s]

RELAXTEVAP = 0.2

Evaporation relaxation time [s]

OAF = 'WALLIS'

Onset of annular flow model [-]

OAFFILMSPLIT = 'RATIO'

Film mass flow rate at onset of annular flow [-]

OAFBASERATIO = 0.5

Base/Film mass ratio at onset of annular flow [-]

OAFENTRAINED = 'RATIO'

Entrained model at onset of annular flow [-]

OAFDROPRATIO = 0.7

Drop/Liquid mass ratio at onset of annular flow [-]

OAFTRANSITION = [0.100.0]

Annular flow transition function parameters (sigmoid width/location wrt OAF) [m]

DEPOSITION = 'GOVAN'

Drop deposition model [-]

ENTRAINMENT = 'GOVAN'

Film entrainment model [-]

MOMENTFILM = 'ALGEBRAIC'

Film momentum conservation model [-]

MOMENTDROP = 'SLIP'

Drop momentum conservation model [-]

DROPSLIP = 1.0

Drop velocity ratio [-]

THINFILMFRIC = 'TURBULENT'

Thin film wall friction model [-]

THINFILMTHICK = 1E-4

Thin film thickness [m]

VAPORFRIC = 'WALLIS'

Vapor friction model [-]

VAPORFRICCST = 0.005

Vapor friction constant [-]

POSFILM = true
 Keep positive film flowrate/thickness
DROPDIAM = 1E-3
 Drop diameter [mm]
DROPDRAG = 'CONSTANT'
 Drop drag model
DROPDRAGCOEF = 0.45
 Drop drag coefficient
BASEEQTHICKCOEF = [5.37E-5-0.641.21]
 Base equilibrium thickness coefficient
RELAXTB = 0.2
 Base film relaxation time
MOMENTBASE = 'ALGEBRAIC'
 Base Film momentum conservation model [-]
SHAPEFACTORCOEF = [1.325E52]
 Wave shape factor coefficients
EQSTROUHALCOEF = [1.1236E-40.5]
 Wave equilibrium Strouhal coefficients
WAVEDRAGCOEF = [0.021.350E50.437]
 Wave drag coefficient
WAVEFREQUENCY = 'RELAXATION'
 Wave number conservation model
RELAXTW = 0.2
 Wave relaxation time
MOMENTWAVE = 'FULL'
 Film momentum conservation model [-]
WAVEMIXCOEF = 0.0
 Wave mixing coefficient
G = 9.81
 [m/s²] Gravitational acceleration

class Inputs.Geometry(*filePath, geometryID*)

Bases: [Inputs.Input](#)

GEOMETRY Summary of this class goes here Detailed explanation goes here

Constructor Summary

Geometry(*filePath, geometryID*)

MODEL Construct an instance of this class Detailed explanation goes here

Property Summary

ID

Channel *ID*

LENGTH = 1

Axial length [m]

AREA = 1

Coolant area [m²]

PERIM = 1

Perimeters [m]

ANGLE = 0

Angle [rad]

Method Summary

HDIAM()

HDIAM() Hydraulic diameter

NWALL()

NWALL() Number of walls

class Inputs.**Input**(*inputFilePath*, *key*, *val*)

Bases: dynamicprops

INPUT Su Detailed explanation goes here

Constructor Summary

Input(*inputFilePath*, *key*, *val*)

INPUT Parse inputFile to construct this class Detailed explanation goes here

Method Summary

validateInputEntry(*objPropname*, *opts*)

VALIDATEINPUTENTRY Description

listInputProperties(*opts*)

LISTINPUTPROPERTIES List of protected obj property names

static convert2JSON(*inputFilePath*)

Read inputFilePath

class Inputs.**FluidProperties**(*P*, *modelObj*)

FLUIDPROPERTIES Summary of this class goes here Detailed explanation goes here

Constructor Summary

FluidProperties(*P*, *modelObj*)

FLUIDPROPERTIES Construct an instance of this class Detailed explanation goes here

Property Summary

FLUID = 'WATER'

Fluid ID

PROPERTIES = 'SATURATED'
 Fluid property assumptions

PRESSURE = 1
 [Pa] System pressure

TSAT = 1
 [K] Saturated fluid temperature

RHOF = 1
 [kg/m³] Saturated liquid mass density

RHOG = 1
 [kg/m³] Saturated vapor mass density

MUF = 1
 [Pa.s] Saturated liquid viscosity

MUG = 1
 [Pa.s] Saturated vapor viscosity

HF = 1
 [J/kg] Saturated liquid enthalpy

HG = 1
 [J/kg] Saturated vapor enthalpy

HFG = 0
 [J/kg] Latent heat of evaporation

SIGMA = 1
 [N/m] Surface tension

KF = 1
 [W/m/K] Saturated liquid conductivity

KG = 1
 [W/m/K] Saturated vapor conductivity

PRANDTLF = 1
 [-] Saturated liquid Prandtl number

PRANDTLG = 1
 [-] Saturated vapor Prandtl number

Method Summary

T(H)
T() Fluid temperature [K] at obj.PRESSURE and H

H(T)
H() Fluid enthalpy [J/kg] at obj.PRESSURE and T

RHOL(H)
RHOL() Liquid mass density (subcooled to saturated) [kg/m³]

RHOV(H)
RHOV() Vapor mass density (saturated to superheated) [kg/m³]

MUL(*H*)
 MUL() Liquid dynamic viscosity [Pa-s]

MUV(*H*)
 MUV() Vapor dynamic viscosity [Pa-s]

KL(*H*)
 KL() Liquid conductivity (subcooled to saturated) [W/m/K]

KV(*H*)
 KV() Vapor conductivity (saturated to superheated) [W/m/K]

PRANDTLL(*H*)
 PRANDTLL() Liquid Prandtl number (subcooled to saturated) [-]

PRANDTLV(*H*)
 PRANDTLV() Vapor Prandtl number (saturated to superheated) [-]

plot(*H*)
 PLOT Plot properties for given enthalpy vector

3.3 Session

class Session.**Log**(*LOGMODE*, *opts*)

Bases: handle

LOG Console log manager Detailed explanation goes here

Constructor Summary

Log(*LOGMODE*, *opts*)

LOG Construct an instance of this class Detailed explanation goes here

Property Summary

LOGFID = -1

log file

Method Summary

log(*varargin*)

LOG Log events

diaryOff()

DIARYOFF Turn diary off

diaryOn()

DIARYON Turn diary on

delete()

DELETE Deconstructor of Log

openLog(*opts*)

SETUPLOG Open/Setup the logging file

closeLog()

CLOSELOG Close log file reference

class Session.LogMode

Bases: uint16

LOGMODE Enumeration class of possible log modes

Session.isLegalPath(pathname)

ISLEGALPATH Determine if a path is legal in current OS

class Session.Session(opt)

Bases: handle

SESSION Summary of this class goes here Detailed explanation goes here

Constructor Summary

Session(opt)

SESSION Construct an instance of this class Detailed explanation goes here

Method Summary

setupLog(session, LOGMODE, opts)

SETUPLOG Setup log

makeSessionDirectory()

MAKESESSIONDIRECTORY Make the session directory

Check if session directory is legal and/or exists

3.4 Solvers

3.4.1 FourField Solver

FourField Solver is cooler.

class Solvers.FourField.Base(film)

Bases: [Solvers.AbstractFilm](#)

BASE Summary of this class goes here Detailed explanation goes here

Constructor Summary

Base(film)

BASE Creates a [Base\(\)](#), base Detailed explanation goes here

Property Summary

NZ = 0

% Solver properties

NTIME = 0

[-] Number of time steps

TIME = 0

[s] Time series

DT = 0
[s] Time step size

TIDX = 1
[-] Time step index

Z = 1.
[m] Elevation

HFLUX = 1.
[W/m^2] Film heat flux

W = 1.
Flow properties

U = 1.
[m/s] Velocity

H = 1E6
[J/kg] Enthalpy

ITR
% Iteration properties

Method Summary

EPSILON(*base, zIdx*)
BETA Base film mass flow fraction

BETA(*base, zIdx*)
BETA() Base film interfacial fraction

BETAP(*base, zIdx*)
BETAP() Base film heat flux fraction

ETA(*base, zIdx*)
ETA() Base film deposition fraction

MENT(*base, zIdx*)
MENT() Base entrainment mass flux TODO: use coefficient later

MEVAP(*base, zIdx*)
MEVAP() Base evaporation mass flux

MTURB(*base, zIdx*)
MTURB() Turbulent mass exchange

MWAVE(*base, drop, zIdx*)
MWAVE() Net Mass flux interaction with wave Defined as a source term, Mwave is the

MTOT(*base, drop, zIdx*)
MTOT() Total mass flux of the base film

FWAVE(*base, zIdx*)
FWAVE() Wave interfacial force

FWAVEMASS(*base, drop, zIdx*)
FWAVEMASS() Base mass exchange force

FBASEVAPOR(*base, zIdx*)

FBASEVAPOR() Film base vapor shear stress TODO: consider different way of doing this...

FDEP(*base, drop, zIdx*)

FWAVE Droplet

FTOT(*base, drop, zIdx*)

FTOT() Total

EQTHICK(*base, zIdx*)

EQTHICK() Base equilibrium thickness

TBASE(*base, zIdx*)

TBASE() Base film exposed time The time in which the base film is exposed to the vapor (eq.24)

TDRY(*base, drop, zIdx*)

TDRY() Time duration in which the base is expected to dry out. Not included in the reference paper.

FDRY(*base, drop, zIdx*)

FDRY() Dry fraction

WMIN(*base, drop, zIdx*)

WMIN() Minimum base film mass flow rate (eq. 27) Base film mass flow rate at the end of the intermittent exposed time (TBASE).

WMINL(*base, drop, zIdx*)

WMINL() Minimum base film mass flow rate per perimeter

THICKMIN(*base, drop, zIdx*)

THICKMIN() Minimum base film thickness (eq. 28)

copyFlowProperties(*srcObj, targetObj, opts*)

COPYFLOWPROPERTIES

class Solvers.FourField.Drop

Bases: *Solvers.ThreeField.Drop*

DROP Summary of this class goes here Detailed explanation goes here

class Solvers.FourField.Film(*inputSet, fluid*)

Bases: *Solvers.AbstractFilm*

FILM Summary of this class goes here Detailed explanation goes here

Constructor Summary

Film(*inputSet, fluid*)

FILM Creates a *Film()*, film Detailed explanation goes here

Property Summary

NZ = 0

Solver properties

NTIME = 0

[-] Number of time steps

TIME = 0

[s] Time series

DT = 0
[s] Time step size

TIDX = 1
[-] Time step index

Z = 1.
[m] Elevation

HFLUX = 1.
[W/m^2] Film heat flux

MEVAP = -1.
[kg/s/m^2] Evaporation mass flux

ITR
Iteration properties

W
Flow properties

U
) double {mustBeNumeric} = 1. % [m/s] Velocity
Type
(
Type
,

H
) double {mustBeNumeric} = 1E6 % [J/kg] Enthalpy
Type
(
Type
,

Method Summary

distributeOAFW(*film*, *W*, *zIdx*)

DISTRIBUTEOAFW Distributes film mass flow rate at onset of annular flow to base and wave, based on option: OAFFILMSPLIT

EQUIL(*film*, *Wf*, *zIdx*)

EQUIL() calculate base/wave split ratio in equilibrium

initializeBaseAndWave(*film*, *WIN*, *ITR*)

INITIALIZEBASEANDWAVE Initialize base and wave arrays given WTOT

copyFlowProperties(*srcObj*, *targetObj*, *opts*)

COPYFLOWPROPERTIES

class Solvers.FourField.Wave(*film*)

Bases: *Solvers.AbstractFilm*

WAVE Summary of this class goes here Detailed explanation goes here

Constructor Summary

Wave(*film*)

WAVE Creates a [Wave\(\)](#), wave Detailed explanation goes here

Property Summary

NZ = 0

% Solver properties

NTIME = 0

[-] Number of time steps

TIME = 0

[s] Time series

DT = 0

[s] Time step size

TIDX = 1

[-] Time step index

Z = 1.

[m] Elevation

HFLUX = 1.

[W/m^2] Film heat flux

W = 1.

Flow properties

U = 1.

[m/s] Velocity

H = 1E6

[J/kg] Enthalpy

FREQUENCY = 1

[Hz] Wave frequency

ITR

% Iteration properties

Method Summary

WL(*wave, zIdx*)

[WL\(\)](#) wave mass flow rate per unit perimeter

THICK(*wave, zIdx*)

[THICK\(\)](#) wave thickness

BETA(*wave, zIdx*)

[BETA\(\)](#) Wave film interfacial fraction

EPSILON(*wave, zIdx*)

BETA Wave mass flow fraction

BETAP(*wave, zIdx*)

[BETAP\(\)](#) Wave film heat flux fraction

ETA(*wave, zIdx*)
ETA() Wave film deposition fraction

MENT(*wave, zIdx*)
MENT() Wave entrainment mass flux

MEVAP(*wave, zIdx*)
MEVAP() Wave evaporation mass flux

MTURB(*wave, zIdx*)
MTURB() Turbulent mass exchange

MBASE(*wave, drop, zIdx*)
MWAVE Mass flux interaction with base

MTOT(*wave, drop, zIdx*)
MTOT() Total mass flux

FWALL(*wave, zIdx*)
FWALL() Film wall shear stress

FBASE(*wave, zIdx*)
FBASE() Base interfacial force

FBASEMASS(*wave, drop, zIdx*)
FBASEMASS() Wave mass exchange force

FVAPOR(*wave, zIdx*)
FVAPOR() Film vapor shear stress

FDRAG(*wave, zIdx*)
FDRAG() Film vapor shear stress due to drag

FSHEAR(*wave, zIdx*)
FSHEAR() Film vapor shear stress

FTOT(*wave, drop, zIdx*)
FTOT() Total

SHAPEFACTOR(*wave, zIdx*)

EQSTROUHAL(*wave, zIdx*)
STROUHAL Correlation of equilibrium Strouhal number

EQFREQUENCY(*wave, zIdx*)

EQPERIOD(*wave, zIdx*)
EQPERIOD() Inverse of EQFREQ

SPACING(*wave, zIdx*)
SPACING() Wave spacing

PERIOD(*wave, zIdx*)
PERIOD() Wave time period Inverse of wave frequency

N(*wave, zIdx*)
N() Wave number density Inverse of wave spacing

WIDTH(*wave, zIdx*)

WIDTH() Wave width Function of amplitude, Shape factor

AMP(*wave, zIdx*)

AMP() Wave amplitude Function of WL, rho, Shape factor, frequency

DELTA_FREQ(*wave, zIdx*)

DELTA_FREQ() Wave frequency source/sink

DRAG_COEF(*wave, zIdx*)

DRAG_COEF() Wave drag coefficient

3.4.2 Mixture Solver

Mixture Solver is pretty cool.

class Solvers.Mixture.Vapor(*mix*)

Bases: *Solvers.AbstractPhase*

VAPOR Summary of this class goes here Detailed explanation goes here

Constructor Summary

Vapor(*mix*)

VAPOR Construct an instance of this class Detailed explanation goes here

Method Summary

TIME(*vapor*)

TIME() Time series [s] Detailed explanation goes here

Z(*vapor, zIdx*)

Z() Axial nodes [m] Detailed explanation goes here

X(*vapor, zIdx*)

X() Mass fraction [-]

VF(*vapor, zIdx*)

VF() Void fraction [-]

W(*vapor, zIdx*)

W() Mass flow rate [kg/s]

U(*vapor, zIdx*)

U() Velocity [m/s] NOTE: need to be verified

H(*vapor, zIdx*)

H() Enthalpy [J/kg]

MFLUX(*vapor, zIdx*)

MFLUX() Mass flux [kg/m^2-s]

RE(*vapor, zIdx*)

RE() Reynolds number [-]

class Solvers.Mixture.**Mixture**(*inputSet, fluid*)

Bases: [Solvers.AbstractField](#)

MIXTURE Summary of this class goes here Detailed explanation goes here

Constructor Summary

Mixture(*inputSet, fluid*)

MIXTURE Creates a [Mixture\(\)](#), mix Detailed explanation goes here

Property Summary

NZ = 0

Solver properties

NTIME = 0

[-] Number of time steps

TIME = 0

[s] Time series

DT = 0

[s] Time step size

TIDX = 1

[-] Time step index

Z = 1.

[m] Elevation

HFLUX = 1.

[W/m^2] Wall heat flux

W = 1.

Flow properties

P = 7E6

[Pa] Pressure

H = 1E6

[J/kg] Enthalpy

DP

[-] Detailed pressure drops

ITR

Iteration properties

liquid

Phases

DZ = 0

[m] Axial step size

Method Summary

MFLUX(*mix, zIdx*)

[MFLUX\(\)](#) Mass flux [kg/m^2-s]

XEQ(*mix, zIdx*)

XEQ() Equilibrium quality [-] This function only retrieves the *mix.xeq* values pre-calculated when *mix.H* is set. This is to eliminate the redundant calculation as a result of requent function calls. The values are calculated via *mix.XEQ_CALC()*

X(*mix, zIdx*)

X() Vapor quality [-] This function only retrieves the *mix.x* values pre-calculated when *mix.H* is set. This is to eliminate the redundant calculation as a result of requent function calls. The values are calculated via *mix.X_CALC()*

VF(*mix, zIdx*)

VF() Void fraction [-]

RHO(*mix, zIdx*)

RHO() Density [kg/m³]

MU(*mix, zIdx*)

MU() Dynamic viscosity [Pa-s]

U(*mix, zIdx*)

U() Velocity [m/s]

JL(*mix, zIdx*)

JL() Superfacial liquid velocity [m/s]

JG(*mix, zIdx*)

JG() Superfacial vapor velocity [m/s]

RE(*mix, zIdx*)

RE() Reynolds number [-]

REL(*mix, zIdx*)

REL() Liquid-equivalent Reynolds number [-]

FW(*mix, zIdx*)

FW() Wall friction factor [-]

TAUW(*mix, zIdx*)

TAUW() wall shear stress [Pa]

KLOSS(*mix, zIdx*)

KLOSS() Local pressure loss coefficient [-] TODO: NEED TO BE VERIFIED

DPGRAV(*mix, zIdx*)

DPGRAV() DPK Gravitational pressure loss [Pa]

DPWALL(*mix, zIdx*)

DPWALL() Wall friction pressure drop [Pa]

DPACCZ(*mix, zIdx*)

DPACCZ() Spatial acceleration pressure drop [Pa]

DPACCT(*mix, Uold, zIdx*)

DPACCT() Temporal acceleration pressure drop [Pa]

DPK(*mix, zIdx*)

DPK() Local pressure loss [Pa]

DPTOT(*mix, Uold, zIdx*)

DPTOT() Total pressure loss [Pa]

DPPARTS(*mix, Uold, zIdx*)

DPPARTS() Pressure loss components[Pa]

T(*mix, zIdx*)

T() Temperature [K]

OAFIDX(*mix*)

OAFIDX() Onset of annular flow node

OAFZ(*mix*)

OAFZ() Onset of annular flow elevation

OAFWL(*mix*)

OAFWL() Liquid mass flow rate at onset of annular flow

AFFNC(*mix, zIdx*)

AFFNC() Annular flow function

AFDISTR(*mix, param1, param2, zIdx*)

AFDISTR() Annular flow distribution function

class Solvers.Mixture.**Liquid**(*mix*)

Bases: *Solvers.AbstractPhase*

LIQUID Summary of this class goes here Detailed explanation goes here

Constructor Summary

Liquid(*mix*)

LIQUID Construct an instance of this class Detailed explanation goes here

Method Summary

TIME(*liquid*)

TIME() Time series [s] Detailed explanation goes here

Z(*liquid, zIdx*)

Z() Axial nodes [m] Detailed explanation goes here

X(*liquid, zIdx*)

X() Mass fraction [-]

VF(*liquid, zIdx*)

VF() Void fraction [-]

W(*liquid, zIdx*)

W() Mass flow rate [kg/s]

U(*liquid, zIdx*)

U() Velocity [m/s] NOTE: need to be verified

H(*liquid, zIdx*)

H() Enthalpy [J/kg]

MFLUX(*liquid*, *zIdx*)
MFLUX() Mass flux [kg/m²-s]
RE(*liquid*, *zIdx*)
RE() Reynolds number [-]

3.4.3 ThreeField Solver

ThreeField Solver is pretty cool.

class Solvers.ThreeField.**Drop**(*inputSet*, *fluid*)

Bases: *Solvers.AbstractField*

DROP Summary of this class goes here Detailed explanation goes here

Constructor Summary

Drop(*inputSet*, *fluid*)

DROP Creates a *Drop()* ?solver? drop Detailed explanation goes here

Property Summary

NZ = 0

Solver properties

NTIME = 0

[-] Number of time steps

TIME = 0

[s] Time series

DT = 0

[s] Time step size

TIDX = 1

[-] Time step index

Z = 1.

[m] Elevation

W = 1.

Flow properties

U = 1.

[m/s] Velocity

H = 1E6

[J/kg] Enthalpy

ITR

Iteration properties

mix = NaN

Mixture

Method Summary

CONC(*drop*, *zIdx*)

Drop concentration

MDEP(*drop*, *zIdx*)

Drop deposition mass flux

RE(*drop*, *zIdx*)

RE() Reynolds number [-]

DIAM(*drop*, *zIdx*)

DIAM() drop diameter

AREA(*drop*, *zIdx*)

AREA() drop interfacial area

VOLUME(*drop*, *zIdx*)

VOLUME() drop volume

DENSITY(*drop*, *zIdx*)

DENSITY() drop number density

AI(*drop*, *zIdx*)

AI() drop volumetric interfacial area

DRAG(*drop*, *zIdx*)

DRAG() drop drag coefficient

FBUOY(*drop*, *zIdx*)

FBUOY() Drop buoyancy

FGRAV(*drop*, *zIdx*)

FGRAV() Drop gravity

FDRAG(*drop*, *zIdx*)

FVAPOR drop vapor drag

FENT(*drop*, *film*, *zIdx*)

FENT() Film entrainment shear

FTOT(*drop*, *film*, *zIdx*, *opt*)

FTOT() Total

USLIP(*drop*, *zIdx*)

Slip drop velocity model

UALGEBR(*drop*, *film*, *zIdx*)

Algebraic drop velocity model (consistent with mixture model)

UEQUIL(*drop*, *film*, *zIdx*, *opt*)

UEQUIL() Drop velocity based on equilibrium model (Ftot = 0)

class Solvers.ThreeField.Film

Bases: *Solvers.AbstractFilm*

FILM Summary of this class goes here Detailed explanation goes here

Property Summary

NZ = 0
 Solver properties
NTIME = 0
 [-] Number of time steps
TIME = 0
 [s] Time series
DT = 0
 [s] Time step size
TIDX = 1
 [-] Time step index
Z = 1.
 [m] Elevation
HFLUX = 1.
 [W/m²] Film heat flux
MEVAP = -1.
 [kg/s/m²] Evaporation mass flux
W = 1.
 Flow properties
U = 1.
 [m/s] Velocity
H = 1E6
 [J/kg] Enthalpy
ITR
 Iteration properties

Solvers is pretty cool.

class *Solvers.AbstractFilm*(*inputSet, fluid*)

Bases: *Solvers.AbstractField*

ABSTRACTFILM Summary of this class goes here

Detailed explanation goes here

Constructor Summary

AbstractFilm(*inputSet, fluid*)

ABSFILM Creates an abstract film, absfilm

Detailed explanation goes here

Property Summary

DZ = 0

[m] Axial step size

Method Summary

WL(*absfilm*, *zIdx*)

WL() Film mass flow rate per unit perimeter

THICK(*absfilm*, *zIdx*)

THICK() Film thickness

RE(*absfilm*, *zIdx*)

RE() Film Reynolds number [-]

MENT(*absfilm*, *zIdx*)

MENT() Film entrainment mass flux

MTOT(*absfilm*, *drop*, *zIdx*)

MTOT() Total

CW(*absfilm*, *zIdx*)

CW() Wall friction factor

FWALL(*absfilm*, *zIdx*)

FWALL() Film wall shear stress

UWALL(*absfilm*, *zIdx*)

UWALL() [m/s] Film wall velocity

YPLUS2THICK(*absfilm*, *yplus*, *zIdx*)

YPLUS2THICK() Calculate thickness from wall unit value

CV(*absfilm*, *zIdx*)

CV() Film/vapor interfacial friction factor

FVAPOR(*absfilm*, *zIdx*)

FVAPOR() Film vapor shear stress

FBUOY(*absfilm*, *zIdx*)

FBUOY() Film buoyancy

FGRAV(*absfilm*, *zIdx*)

FGRAV() Film gravity

FDEP(*absfilm*, *drop*, *zIdx*)

FDEP() Drop deposition shear

FTOT(*absfilm*, *drop*, *zIdx*)

FTOT() Total

UALGEBR(*absfilm*, *zIdx*)

UALGEBR() Film velocity based on simple algebraic model

UEQUILS(*absfilm*, *zIdx*)

UEQUILS() Film velocity based on simple equilibrium model ($F_{wall} + F_{vapor} = 0$)

UEQUIL(*absfilm*, *drop*, *zIdx*)

UEQUIL() Film velocity based on complete equilibrium model ($F_{tot} = 0$)

class Solvers.AbstractField

Bases: `matlab.mixin.Copyable`

ABSTRACTFIELD Summary of this class goes here

Detailed explanation goes here

Property Summary**NZ**

[-] Number of axial steps

NTIME

[-] Number of time steps

TIME

[s] Time series

DT

[s] Time step size

TIDX

[-] Time step index

Z

[m] Elevation

ITR

Iteration properties

flowProperties = {'W','U','H'}

Flow properties used for copying

Method Summary**memoizeFunction**(*methodStr, methodHandle, varargin*)

MEMOIZEDMETHOD Implement a mechanism for registering memoizeable functions

Adapted from <https://stackoverflow.com/a/75037451>

For the first call with a particular method, create and memoize a function handle view of the method

struct()STRUCT Converter to [struct\(\)](#)**copyFlowProperties**(*srcObj, targetObj, opts*)

COPYFLOWPROPERTIES

class Solvers.AbstractPhase

Bases: handle

ABSTRACTPHASE Summary of this class goes here Detailed explanation goes here

class Solvers.SolverPlotter(*Titles, WallIdxs*)

Bases: handle

SOLVERPLOTTER Framework for generating solver plots Detailed explanation goes here

Constructor Summary**SolverPlotter**(*Titles, WallIdxs*)

SOLVERPLOTTER Construct an instance of this class Detailed explanation goes here

Property Summary**FontSize = 10**

Font size of text in plots

```
Title = ""  
    Figure title  
WallIdx = 1  
    Wall index  
Grid = 'on'  
    Grid option
```

Method Summary

```
plotz(plotters, YData, fieldName, opts)  
    PLOTZ Summary of this method goes here Detailed explanation goes here  
gca(plotters)  
    Loop through plotters  
ylim(plotters, newLim)  
    Loop through plotters  
setZs(plotters, Zs)  
    Loop through plotters  
legend(plotters, varargin)  
    Loop through plotters  
static fieldName2plotStyle(fieldName)  
    Default colors
```

class Solvers.SolverState

Bases: uint16
SOLVERSTATE Enumeration class of possible solver states

class Solvers.AbstractSolver(*inputSet*)

Bases: handle
ABSTRACTSOLVER Summary of this class goes here Detailed explanation goes here

Constructor Summary

```
AbstractSolver(inputSet)  
    ABSTRACTSOLVER Constructor
```

Property Summary

```
STATE  
    Solver state defined by SolverState enum
```

Method Summary

```
log(solver, varargin)  
    LOG Log events  
static CreateITR(NZ, ITRFields)  
    Create inner iteration value struct
```

EXTERNAL PACKAGES

4.1 CoolPropWrapper

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